

Hasun Lee

Computer Vision Engineer

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Education

M.S. Seoul National University , Mechanical Engineering - Interactive & Networked Robotics Laboratory (INRoL), Advisor Prof. Dongjun Lee - Thesis: Contact Wrench Estimation for Large-Size Operational Aerial Skeleton System with Distributed IMUs	Seoul, Korea Mar 2019 – Aug 2021
B.S. Pohang University of Science and Technology (POSTECH) , Physics - Graduated with Magna Cum Laude - GPA: 3.73/4.3 - Thesis: 3D printed Watt balance device and mass measuring with electro-magnetic force	Pohang, Korea Mar 2014 – Feb 2019
High School Seoul Science High School , Science Diploma Early Entrance Student Body Vice President	Seoul, Korea Mar 2011 – Feb 2014

Experience

StradVision Corp. , Computer Vision Engineer Joined prior to the company's KOSDAQ IPO (expected Jun. 2026); contributing through pre-IPO growth phase.	Seoul, Korea Sept 2021 – present 4 years 10 months
- LSS-based 3D detector for multi-fisheye cameras on a confirmed production project - Set up and modified an LSS/BEVDet training framework for multi-fisheye inputs - Built a 3D Asset Harvesting pipeline combining 2D/3D generation and segmentation models; lifted production-homologation Recall for parking stoppers and pedestrians from 0.20 to 0.997 - Applied data augmentation to real RGB images using Depth-Pro and SAM2 - Developed a fisheye image synthesis and data acquisition pipeline in the CARLA simulator	

Publications

Wrench Estimation of Modular Manipulator with External Actuation and Joint Locking Y. Kim, Hasun Lee , J. Lee, Dongjun Lee (IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS))	Oct 2023
Development of Downsized LASDRA with 2-DoF Joint Locking Device Hasun Lee , Y. Kim, H. Yang, Dongjun Lee (Aerial Robotic Systems Physically Interacting with the Environment (AIRPHARO))	Oct 2021
Preliminary Results on Downsized LASDRA System with Joint Locking Hasun Lee , Y. Kim, H. Yang, Dongjun Lee (Ubiquitous Robots (UR))	July 2021
Robust Motion Control of Robotic Systems with Environmental Interaction via Data-Driven Inversion of CPG S. Park, Hasun Lee , Dongjun Lee (International Conference on Control, Automation and Systems (ICCAS))	Oct 2020

Preliminary Results on Motion Control of Robotic Systems with Environmental Interaction via Data-Driven Inversion of CPG

Aug 2020

S. Park, **Hasun Lee**, Dongjun Lee

(Korea Robotics Society Annual Conference (KRoC))

Variable Speed Control Moment Gyro's Gimbal Arrangement Optimization and Control Allocation Improving

July 2020

Hasun Lee, S. Park, Dongjun Lee

(ICROS Annual Conference)

Projects

Auto Parking Software Mass Production Development

Sept 2021 – present

StradVision — parking space recognition and obstacle detection for automatic parking.

- Main contributor for the object detection feature

Very Long, Slender and Light Robot Arm with Distributed Rotor Actuation

Apr 2020 – Mar 2021

National Research Foundation of Korea — aerial operation and manipulation.

- Development of joint locking LASDRA system
- External wrench estimation with multiple IMU sensor fusion using UKF

Attitude Control Algorithm for Satellite with CMG Singularity Avoidance

Apr 2019 – Mar 2020

Justek Inc.

- Newton-Euler dynamic analysis and control of variable speed CMG
- Design optimization of CMG cluster by data-driven method

Teaching Experience

Seoul National University, Teaching Assistant

Seoul, Korea

Undergraduate Course — Mechanical System Modeling and Control (Spring 2020)

Mar 2020 – June 2020

Awards

1st Prize — International Astronomy Olympiad (IAO)

Ukraine, Kazakhstan

2010 and 2011

2011

National Science & Technology Scholarship (NSTS)

Korea Student Aid Founda-

2014–2018

tion

2014

Extracurricular Activity

Financial Engineering Intern — NH Investment & Securities

July 2017

Price determining of ELS products by Monte-Carlo method

Exchange Student — National University of Singapore

Singapore

One semester

2016

Skills

Programming: Python, C/C++, Matlab, ROS, LabVIEW

Libraries: PyTorch, OpenCV, NumPy, ONNX

DevOps: Git, GitHub, Docker

Simulators: CARLA, Gazebo

Design: SolidWorks, Onshape

Operating Systems: Linux, Windows

Languages

Korean: Native

English: Professional working proficiency